Harsh Tripathi

Portfolio | +91-8279829431 | harshtripathi803@gmail.com | GitHub | LinkedIn

Professional Summary

Data-focused Python Developer and aspiring Data Analyst with hands-on experience in data processing, machine learning, and automation. Skilled in data cleaning, exploratory data analysis, and building predictive models. Experienced in transforming raw data into actionable insights and developing data-driven solutions using Python and modern tools..

Experience

Python Developer Intern –Ishan Ceramics Private Limited, Agra (Oct. 2025 – Mar. 2026)

- Developed **data-driven automation solutions** using Python to streamline business workflows.
- Designed **AI-based agents** to support data-backed decision-making and improve operational efficiency.
- Built **image processing pipelines** using OpenCV and NumPy for structured data extraction.
- Integrated data workflows with **ERP systems and databases**, ensuring accurate data flow.
- Analyzed business requirements and converted them into **data-driven insights and automation solutions**.
- Improved system performance through debugging and optimization.

Full Stack Web Developer Intern – Infoseeds, Delhi (Jun. 2024 – Aug. 2024)

- Improved frontend performance by optimizing components for **responsiveness and cross-browser compatibility**.
- Conducted **UI testing and bug reporting**, ensuring high-quality user experience across applications..
- Validated and tested **REST API integrations**, ensuring correct data communication between frontend and backend.
- Collaborated with development teams to **identify and fix UI/UX issues**, improving usability.
- Assisted in improving overall application quality through **structured testing and debugging practices**.

Projects

Monkeypox Detection System (Machine Learning Project)

- Developed ML models using **Xception, EfficientNet, ResNet, MobileNet**.
- Applied **ensemble learning techniques** to improve prediction accuracy.
- Performed **data preprocessing, augmentation, and feature engineering**.
- Evaluated models using Accuracy, Precision, Recall, and F1-score.

Air Quality Index Prediction System

- Built predictive models using **Linear Regression, Random Forest, XGBoost, ANN**.
- Performed **EDA, data cleaning, and feature engineering**.
- Visualized insights using Matplotlib and Seaborn.
- Evaluated performance using **R^2, MAE, RMSE**.

Django StudyBuddy Application

- Developed and tested chat room creation and authentication workflows using Django REST Framework.
- Implemented input validation and error handling to ensure secure and reliable messaging.
- Integrated APIs with frontend clients for real-time communication and smooth user experience.

Technical Skills & Tools

Programming: Python, SQL with strong experience in data processing and scripting
Data Analysis: EDA, data cleaning, and visualization using Pandas, NumPy, Matplotlib, Seaborn
Machine Learning: Regression, classification, clustering with model evaluation and tuning
Libraries & Tools: Scikit-learn, TensorFlow, Keras, OpenCV, Jupyter Notebook
Technologies: Power BI, Tableau, Excel, Git, REST APIs, MySQL, MongoDB, PostgreSQL

Education

B.Tech – Information Technology (CGPA: 6.4) (2021–2025)
Hindustan College of Science and Technology, Mathura

Senior Secondary Education (ISC Board) (2019–2020)
Ragendra Swarup Public School , Agra